

MANIPAL FORMAT | PRACTICE PAPER
THIRD SEMESTER B.TECH. | MIDSEM PRACTICE
DATA ANALYTICS [CSS 2103]
COMPUTER SCIENCE AND ENGINEERING
SET A: ENSEMBLE OF PAST-PAPER QUESTIONS

Marks: 30**Suggested duration: 90 minutes**

Answer all questions. Questions 1–5 are MCQs worth 1 mark each. Select one correct option per MCQ. The remaining questions are theory questions worth 25 marks in total; show working and draw labelled diagrams where appropriate. Modules 1-3. Independent practice paper; not an official university examination.

Non-programmable calculator permitted. Use the critical values supplied.

Section A: Multiple-choice questions (5 marks)

- 1) A manufacturer claims a population mean accuracy of 0.95. Which null hypothesis tests this claim? (1)
(A) $\mu = 0.95$
(B) $\bar{x} = 0.95$
(C) $\mu \neq 0.95$
(D) $\mu < 0.95$
- 2) A population has known standard deviation 0.02. For an independent sample of size 36, what is the standard error of the sample mean? (1)
(A) 0.12
(B) 0.02
(C) $0.02/6$
(D) $0.02/36$
- 3) With $\bar{x} = 0.947$, $\mu_0 = 0.95$, $\sigma = 0.02$ and $n = 36$, what is the z statistic? (1)
(A) 0.90
(B) -0.90
(C) -9.00
(D) 9.00
- 4) Observations range from 180 to 480. What is the width of each of three equal-width bins? (1)
(A) 60
(B) 150
(C) 300
(D) 100
- 5) Twelve observations are divided into three equal-frequency bins. How many observations are in each bin? (1)
(A) 3
(B) 4
(C) 6
(D) 12

Section B: Theory questions (25 marks)

- 6) (i) Find the standard deviation and skewness for: 78, 91, 68, 54, 82, 47, 60, 78, 97, 58, 62, 50, 50, 63, 92, 75, 79, 63, 42, 61. (5)
(ii) Give an example of the combining-variables concept in dataset preparation.
(iii) A CMO compares three new cola flavours using ratings from 12 customers on a 1-9 scale. Conduct the appropriate analysis. Ratings: Flavour A: 4, 2, 4, 2; Flavour B: 6, 4, 5, 5; Flavour C: 8, 6, 8, 6. Use population standard deviation and Pearson's second skewness coefficient. For (iii), assume independent normal samples with equal variances; use a one-way ANOVA at 5%, with critical $F(2,9) = 4.256$.
- 7) (i) Find the elements of a box plot for: 78, 91, 68, 54, 82, 47, 60, 78, 97, 58, 62, 50, 50, 63, 92, 75, 79, 63, 42, 61. (5)
(ii) Why is it sometimes necessary to create subsets from a large dataset before analysis?
(iii) Measure the strength of the relationship between the rankings assigned by two interviewers to candidates A-L. Interviewer 1: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12. Interviewer 2: 1, 2, 4, 3, 6, 5, 8, 7, 10, 9, 12, 11. For quartiles, use medians of the two halves. Use Spearman rank correlation in (iii).
- 8) A hospital laboratory claims that a new COVID testing kit correctly identifies infection 95% of the time. A researcher believes the true accuracy is less than 95%. A sample of 50 test results has mean accuracy 0.93; the known population standard deviation is 0.04. Test the claim at the 5% significance level. Critical z-value: -1.645. (3)
- 9) Test the effectiveness of vaccination in preventing smallpox at the 5% significance level. (5)
Vaccinated: attacked 31, not attacked 469, total 500.
Not vaccinated: attacked 185, not attacked 1315, total 1500.
Overall: attacked 216, not attacked 1784, total 2000. Use Pearson chi-square without continuity correction; critical value for 1 degree of freedom is 3.841. State association, not a causal conclusion.
- 10) Per-acre wheat production (tons) is recorded for three varieties and four fertilizers, with each combination replicated twice. Test whether variety differences and fertilizer differences are significant at the 5% level. Each pair below gives the two replicates, in fertilizer order F1, F2, F3, F4. (5)
Variety 1: (8,9), (9,8), (6,7), (10,11).
Variety 2: (7,6), (8,9), (5,4), (9,10).
Variety 3: (10,9), (11,12), (8,7), (12,13). Assume independent normal errors with common variance. Include the interaction term. Use critical F values: variety (2,12): 3.885; fertilizer (3,12): 3.490.
- 11) A data scientist is analysing the following cholesterol level (mg/dL) readings of 12 patients: [145, 155, 160, 162, 165, 172, 180, 190, 200, 220, 240, 260]. Apply equal-width and equal-frequency binning with 3 bins. Using histogram analysis, explain which binning approach (width-based or frequency-based) better represents the data distribution and why. (2)

End of question paper